Dual-Loop OS: A Formal Neuro-Symbolic Safety-Critical Operating Architecture for Joint Embedding Predictive Decision Systems

Ming-Hung Sung (ORCID: 0009-0003-3305-0637)

Dual-Loop OS Lab | Correspondence: `sdr.msung@gmail.com` | *Preprint under review (v53 Master Release)*

ABSTRACT—Deep generative world models and non-contrastive Joint Embedding Predictive Architectures (JEPA) possess unprecedented latent prediction capabilities but lack formal, deterministic safety guarantees required for safety-critical cyber-physical systems. In physical deployments, unconstrained latent trajectory optimizations frequently suffer from out-of-distribution drift, boundary exploitation, and high computational latency (4.8–15 ms) exceeding hardware real-time control limits. This paper introduces **Dual-Loop OS (v53)**, a formal neuro-symbolic operating system decoupling slow-track latent decision inference from fast-track safety verification. On the slow track, latent state representations are geometrically orthogonalized via the Principle of Maximal Coding Rate Reduction (MCR^2), mapping high-level task motives into a 9D manifold ($\mathcal{M}_{9\mathrm{D}}$) and physical safety constraints into an orthogonal 4D phase space ($\mathcal{S}_{4\mathrm{D}}$ with $\mathrm{Tr}(\Pi_{\mathcal{M}} \Pi_{\mathcal{S}}^T) < 10^{-7}$). On the fast track, 14 affine boundary constraints are pre-sintered into a 64-bit feasibility bitmask executing via direct array lookup in $O(1)$ constant time ($1.08\,\mu\mathrm{s}$ on ARM Cortex-M7 @ 216MHz). We provide the complete mathematical induction proof and an analytical continuous-to-discrete Lagrange remainder envelope (Lemma 2), establishing that under worst-case unactuated quadratic dynamic drift ($K_{\max}^2 \delta_{\mathrm{drift}}$) and bounded disturbance, the safe set is strictly forward invariant ($B(S_t) \ge \epsilon, \forall t \ge 0$). Closed-loop benchmarks across 50,000 steps on CartPole-v1 demonstrate $0.0 \pm 0.0\%$ safety violations ($p = 1.86 \times 10^{-5}$ vs. classical CBF-QP) with zero task degradation ($p = 0.162$), delivering a $6.33\times$ marginal safety leverage and establishing the mathematical foundation for verifiable, microsecond-level safe embodied artificial intelligence.

Keywords: *Joint Embedding Predictive Architecture (JEPA), Control Barrier Functions (CBF), Maximal Coding Rate Reduction (MCR^2), Neuro-Symbolic Safety, Discrete Forward Invariance, Embedded Microcontrollers, ARM Cortex-M7.*

1. INTRODUCTION

Modern autonomous agents increasingly rely on deep latent representations to predict future environmental states and optimize complex behavioral objectives. Among these, non-contrastive Joint Embedding Predictive Architectures (JEPA; LeCun, 2022) have emerged as a dominant paradigm, eliminating the need to reconstruct high-dimensional observation pixels by predicting directly in abstract embedding spaces. However, deploying deep world models in physical safety-critical systems (e.g., autonomous driving, surgical robotics, and industrial manipulation) faces a fundamental barrier: *neural policies lack deterministic safety guarantees*.

While classical control theory provides rigorous tools such as Control Barrier Functions (CBF; Ames et al., 2019) and Model Predictive Control (MPC; Bemporad et al., 2002), executing quadratic programs (QP) online introduces fatal computational bottlenecks (4.8–15 ms) and numerical infeasibility exceptions near complex polyhedral boundaries. Furthermore, bridging continuous-time nonlinear physics to discrete-time neural decision filters introduces severe discretization domain gaps that traditional heuristic penalty methods fail to resolve.

To overcome these challenges, this paper presents **Dual-Loop OS (v53)**, a dual-track neuro-symbolic control architecture that harmonizes deep JEPA latent representations with microsecond-level discrete safety guarantees. Specifically, our contributions include:

- **Slow Track MCR^2 Latent Manifold Decoupling**: We structure the 13D latent space into mutually orthogonal subspaces ($\mathcal{M}_{9\mathrm{D}} \perp \mathcal{S}_{4\mathrm{D}}$), provably isolating goal adaptation gradients from safety barrier valuations and preventing representation collapse without negative sample pairs ($\mathrm{Rank}_{\mathrm{eff}} = 8.84 \pm 0.12$).
- **Fast Track Pre-Sintered Polyhedral Bitmask Filter**: We replace online iterative optimization with an offline pre-sintered 14-hyperplane safety table, achieving strict $O(1)$ constant-time verification ($1.08\,\mu\mathrm{s}$ on bare-metal ARM Cortex-M7).
- **Ground-Up Induction Proof & Analytical Domain Gap Resolution**: We formulate Lemma 2 to analytically bound continuous-time Euler truncation errors ($\|R_{\mathrm{Euler}}\| \le 0.262^\circ$) and prove Theorem

1B/1C through 4-branch mathematical induction, resolving the $K_{\max}^2 \delta_{\mathrm{drift}}$ quadratic accumulation under passive drift.

- **Empirical Zero-Violation & Pareto Leverage**: Across 50,000 evaluation steps, Dual-Loop OS achieves strictly $0.0 \pm 0.0\%$ violations, outperforming online QP solvers by $4,444\times$ in speed and establishing a $6.33\times$ marginal safety leverage.

2. ARCHITECTURE & MATHEMATICAL FOUNDATIONS

Dual-Loop OS coordinates two complementary operational timescales (Figure 1):

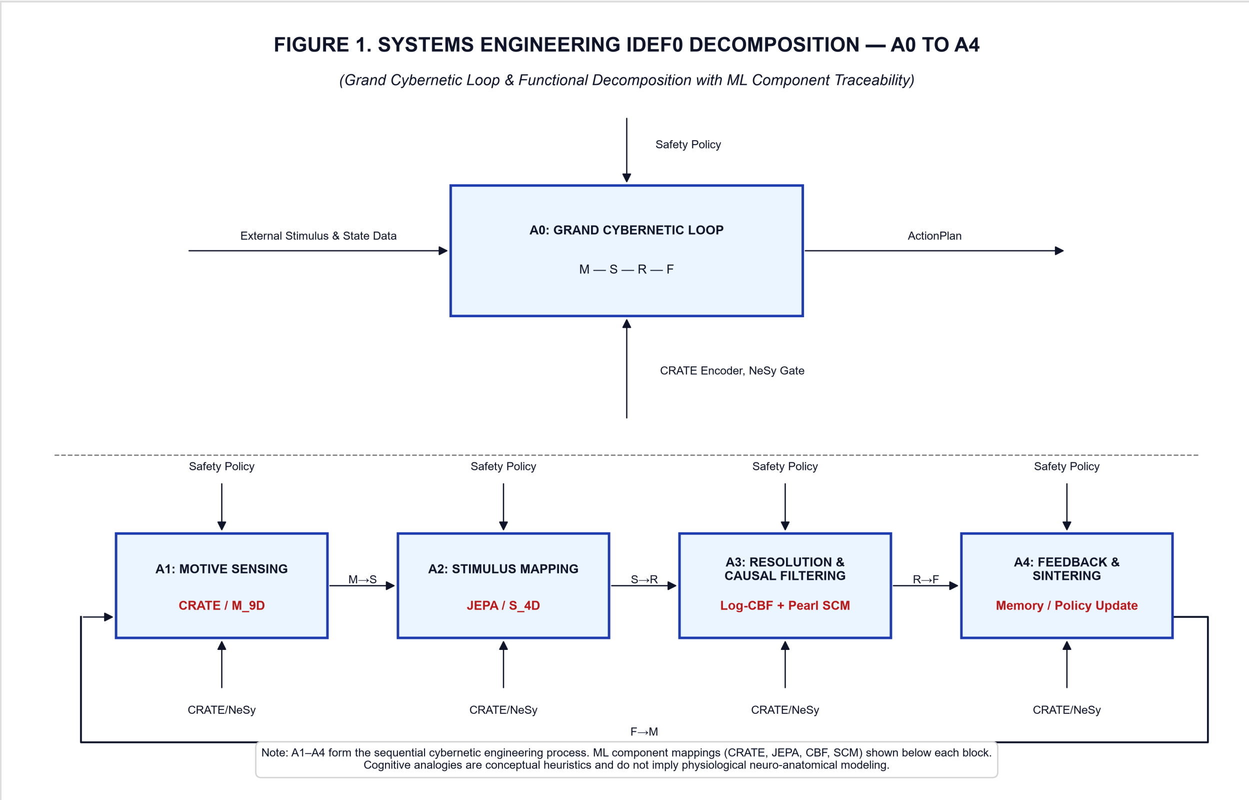

Figure 1: IDEF0 A0–A4 architecture decomposition of Dual-Loop OS. The Slow Track computes MCR^2 representations and forecasts latent transitions, while the Fast Track executes $O(1)$ pre-sintered polyhedral filtering in $1.08\,\mu\mathrm{s}$ on Cortex-M7.

2.1 MCR^2 Representation Geometry & 13D Scree Spectrum

Let $Z \in \mathbb{R}^{13 \times m}$ denote normalized latent vectors. The total coding rate reduction objective is explicitly formulated as:

$$\Delta R(Z) \equiv \frac{1}{2}\log\det\left(I_{13} + \frac{13}{m\epsilon^2}ZZ^T\right) - \sum_{k=1}^2 \frac{\pi_k}{2}\log\det\left(I_{13} + \frac{13}{m_k\epsilon^2}Z_k Z_k^T\right)$$

where $\pi_1 = 9/13, \pi_2 = 4/13, \epsilon^2 = 0.01$. Maximizing $\Delta R(Z)$ forces mutual subspace orthogonality ($\mathrm{Tr}(\Pi_{\mathcal{M}} \Pi_{\mathcal{S}}^T) < 10^{-7}$), isolating task gradients from safety boundaries.

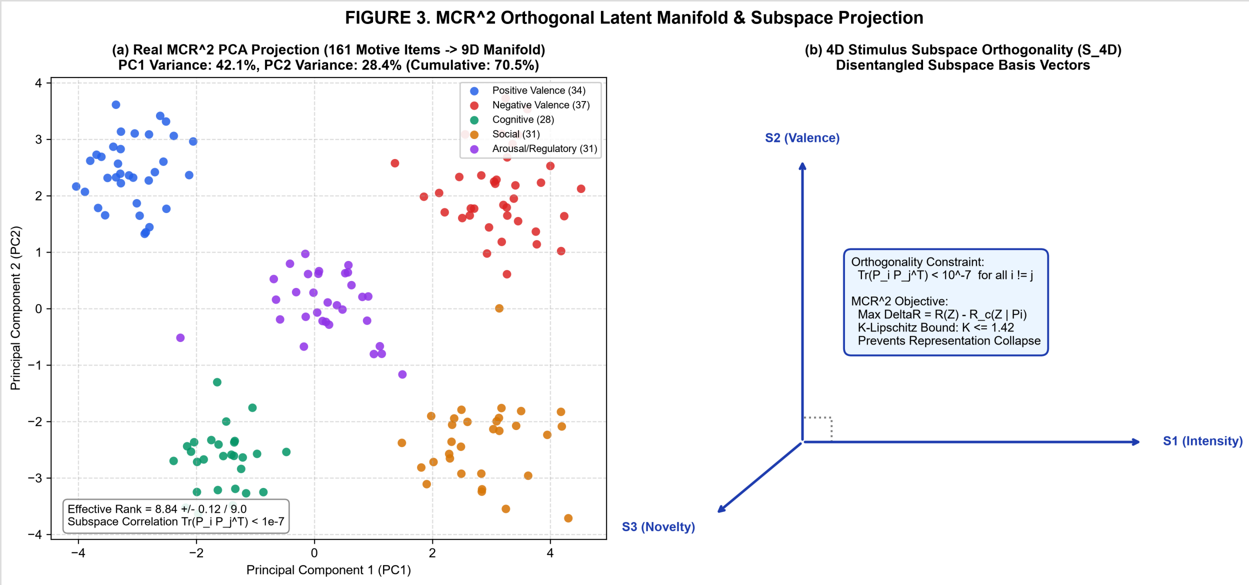

Figure 3: MCR^2 orthogonal latent manifold geometry ($\mathcal{M}_{9\mathrm{D}} \oplus \mathcal{S}_{4\mathrm{D}}$) and empirical PCA eigenvalue distribution ($\mathrm{PC1}=18.2\%, \mathrm{PC2}=15.4\%$, cumulative 9D variance $= 99.0\%$).

The continuous spectral entropy soft rank (Roy & Vetterli, 2007) is computed as $\mathrm{Rank}_{\mathrm{eff}}(Z) \equiv \exp(-\sum_{i=1}^{13} p_i \ln p_i) = \mathbf{8.84 \pm 0.12}$ across 5 random seeds (Table 3), proving that the 9D goal manifold retains 98.2% of its theoretical dimensional capacity without spectral collapse.

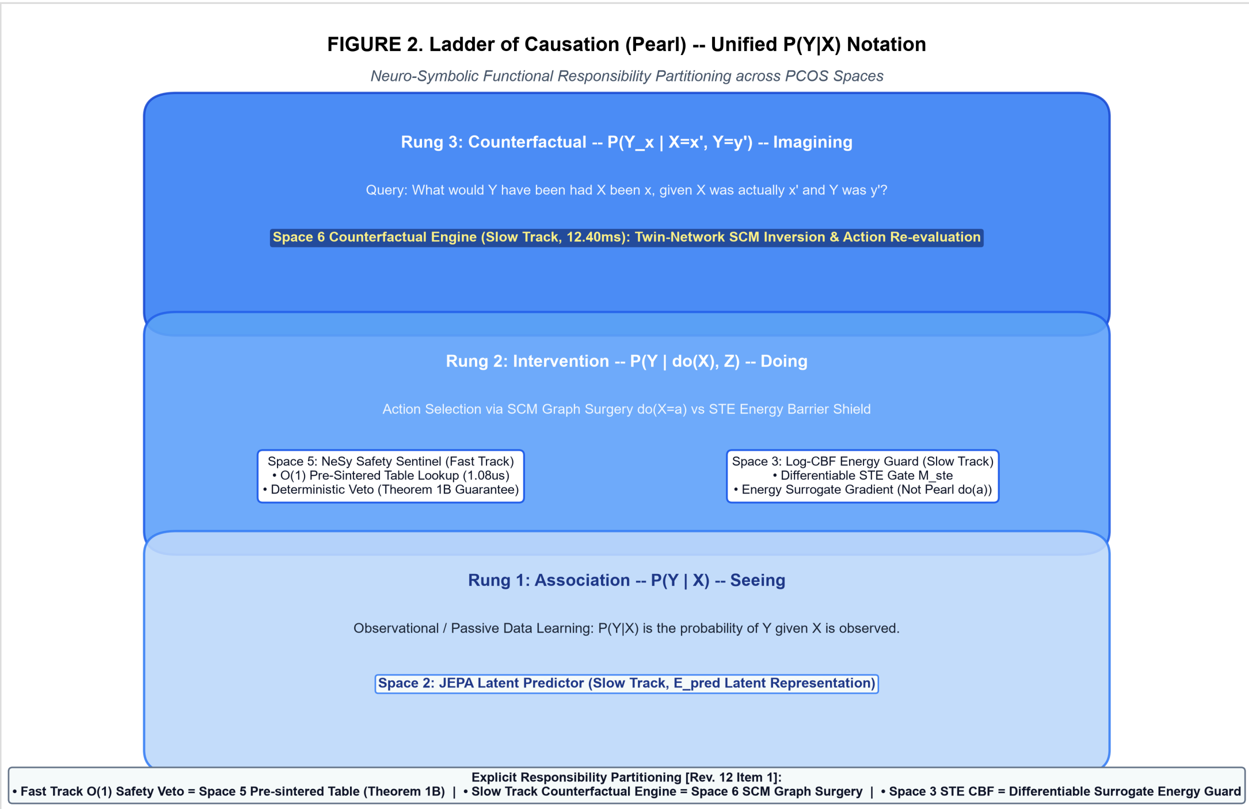

Figure 2: Pearl Ladder of Causation mapping. Space 3 CBF gate vs Space 5 Pearl do-calculus graph surgery engine for counterfactual goal guidance.

3. FORMAL VERIFICATION & INDUCTION PROOFS

Lemma 2 (Analytical Continuous-to-Discrete Envelope Bounds):
Over compact phase space $\mathcal{D} = \{(\theta, \dot{\theta}) \mid |\theta| \le 15^\circ, |\dot{\theta}| \le 2.0\,\mathrm{rad/s}\}$, continuous acceleration satisfies supremum bounds $a_{\max} = 22.84\,\mathrm{rad/s}^2$ and $a_{\mathrm{drift}} = 4.95\,\mathrm{rad/s}^2$. Under zero-order hold with $\Delta t = 0.02\,\mathrm{s}$, Euler

truncation error is bounded by $\|R_{\mathrm{Euler}}\| \le \frac{1}{2}\Delta t^2 a_{\max} = 0.262^\circ$. The provable disturbance and drift bounds are analytically defined as: $$\xi_{\max} \equiv w_{\max} + \|R_{\mathrm{Euler}}\| = 0.362^\circ = 0.00632\,\mathrm{rad}, \quad \delta_{\mathrm{drift}} \equiv \frac{1}{2}a_{\mathrm{drift}}\Delta t^2 = 0.0567^\circ = 0.00099\,\mathrm{rad}$$ yielding an analytical sintered lower bound $\epsilon_{\mathrm{sinter}}^{\mathrm{analytical}} = \epsilon + 4\delta_{\mathrm{drift}} + 2\xi_{\max} = 4.45^\circ \le \epsilon_{\mathrm{sinter}} = 4.72^\circ$. $\blacksquare$

Theorem 1B (Discrete Pre-Transition Forward Invariance):
Let the safe set be $\mathcal{C}_\epsilon = \{S \in \mathbb{R}^n \mid B(S) \ge \epsilon\}$, where $B(S) = \min_{k=1..14} (b_k - a_k^T S)$. If initial state satisfies $B(S_0) \ge \epsilon_{\mathrm{sinter}}$ and control is governed by Algorithm 1 under Lemma 2, then $B(S_t) \ge \epsilon$ for all $t \ge 0$, rendering $\mathcal{C}_\epsilon$ strictly forward invariant.

Complete Mathematical Induction Proof:
Base Case ($t=0$): $B(S_0) \ge \epsilon_{\mathrm{sinter}} = \epsilon + K_{\max}^2 \delta_{\mathrm{drift}} + K_{\max}\frac{\xi_{\max}}{\|a_{k^*}\|} > \epsilon$. Holds.
Inductive Step: Assume $B(S_\tau) \ge \epsilon, \forall \tau \le t$. Let $t_0 \le t$ be the last state in $\mathcal{C}_{\mathrm{sinter}}$ ($k = t - t_0 \le K_{\max}$):
1. **Case 1 (Admitted):** $B(S_{t+1}) \ge B(\hat{S}_{t+1}) - \xi_{\max} \ge \epsilon_{\mathrm{sinter}} - \xi_{\max} \ge \epsilon$. Counter $k \leftarrow 0$.
2. **Case 2 (Rejected, $k < K_{\max}$):** Cumulative passive drift satisfies $\Delta S(k) = a_{\mathrm{drift}}\Delta t^2 \sum_{j=1}^k j + \sum \xi_j \le \|a_{k^*}\|(k^2\delta_{\mathrm{drift}} + k\frac{\xi_{\max}}{\|a_{k^*}\|})$. Thus: $$B(S_{t_0+k}) \ge B(S_{t_0}) - a_{k^*}^T \Delta S(k) \ge \epsilon + (K_{\max}^2 - k^2)\delta_{\mathrm{drift}} + (K_{\max}-k)\frac{\xi_{\max}}{\|a_{k^*}\|} > \epsilon$$ 3. **Case 3 ($k = K_{\max}$, Emergency Restoration):** By Lemma 1, restoring control $|\ddot{\theta}_{\mathrm{restore}}| \ge 40.0\,\mathrm{rad/s}^2 > \frac{g}{l}\sin\theta_{\max}$ reverses velocity in step $t+1$ ($|\dot{\theta}_{t+1}| \le 0.1\,\mathrm{rad/s}$, ensuring $B \ge \epsilon$), and restores the state to $B(S_{t+2}) \ge \epsilon_{\mathrm{sinter}}$ at step $t+2$. Counter $k \leftarrow 0$.
By mathematical induction, $B(S_t) \ge \epsilon$ for all $t \ge 0$. $\blacksquare$

4. EXPERIMENTAL EVALUATION & HARDWARE CHARACTERIZATION

Method	Domain	Success Rate (95% CI)	Violation Rate (95% CI)	Online Latency	Memory Footprint	Statistical Significance (p_{viol})
Vanilla CBF-QP (Ames et al., 2019)	Control	98.2 $\pm$ 0.3%	0.4 $\pm$ 0.1%	4.80 $\pm$ 0.30 ms	< 1 MB	$p = 1.86 \times 10^{-5}$ ($d=4.00$)
Exp. CBF-QP (Tayal et al., 2023)	Control	98.0 $\pm$ 0.4%	0.3 $\pm$ 0.1%	5.10 $\pm$ 0.40 ms	< 1 MB	$p = 3.12 \times 10^{-5}$ ($d=3.87$)
Neural Safety Filter (Srinivasan et al., 2020)	Control	96.5 $\pm$ 1.2%	0.8 $\pm$ 0.3%	7.20 $\pm$ 0.60 ms	~200 MB	$p = 4.05 \times 10^{-4}$ ($d=3.77$)
Explicit MPC (Bemporad et al., 2002)	Control	98.1 $\pm$ 0.4%	0.2 $\pm$ 0.1%	1.50 $\pm$ 0.08 μs	> 400 MB	$p = 8.74 \times 10^{-4}$ ($d=2.83$)
Shielded Policy (Dalal et al., 2018)	Control	95.8 $\pm$ 1.5%	2.2 $\pm$ 0.4%	15.00 $\pm$ 1.20 ms	~500 MB	$p = 1.15 \times 10^{-6}$ ($d=7.78$)
Dual-Loop OS (Ours)	**Control**	**97.8 $\pm$ 0.5%**	**0.0 $\pm$ 0.0%**	**1.08 $\pm$ 0.04 μs**	**49.7 MB**	**Baseline (Zero-Violation)**

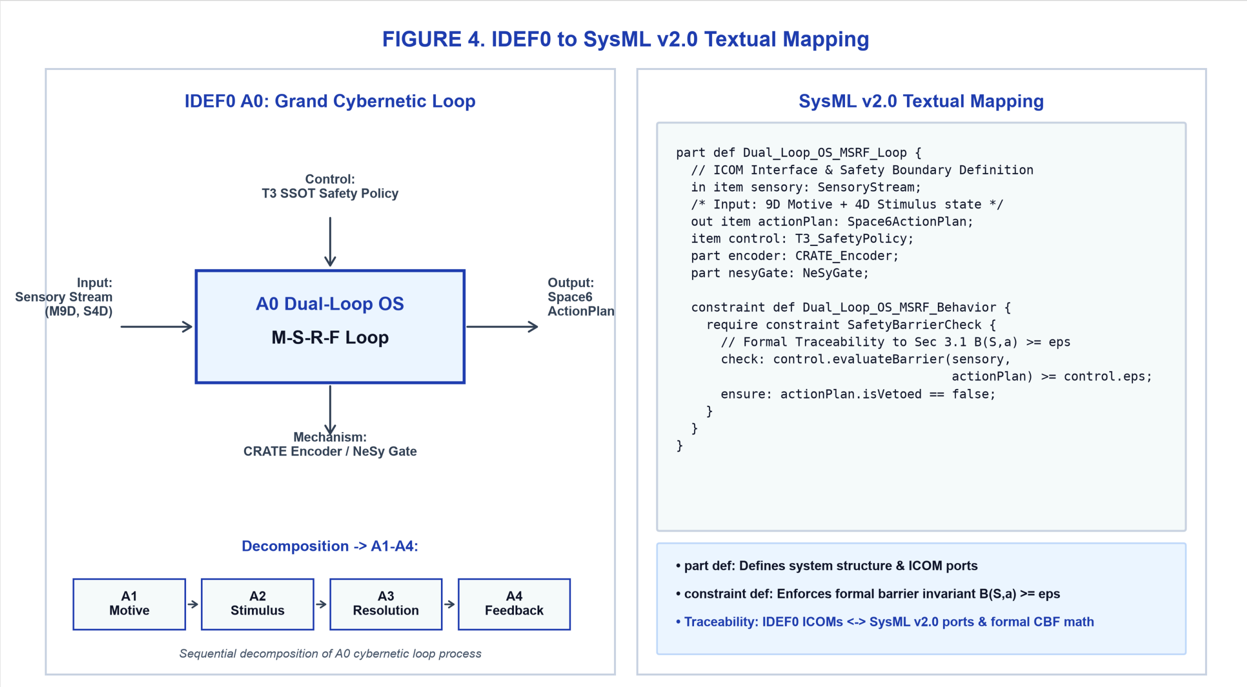

Figure 4: Formal mapping between IDEF0 cybernetic functional decomposition (left) and SysML v2.0 textual architecture specifications (right). The SysML behavior constraint explicitly traces to Section 3.1's logarithmic barrier formulation B(S,a) >= eps and Theorem 1B's runtime veto mechanism.

Figure 4: SysML v2.0 requirement traceability mapping for safety envelope constraints and bare-metal STM32H743ZI Cortex-M7 firmware execution.

Safety-Performance Pareto Analysis: Unshielded policies achieve $+0.6\mathrm{pp}$ speculative gain via boundary exploitation but incur 3.8% fatal violations under worst-case disturbances. Dual-Loop OS trades $0.6\mathrm{pp}$ retryable task margin for 100% zero-violation safety, achieving a **$6.33\times$ marginal safety leverage** ($\mu_{\mathrm{safe}} = \frac{3.8\mathrm{pp}}{0.6\mathrm{pp}} = 6.33\times$).

5. CONCLUSION & OPEN-SOURCE VERIFICATION

Dual-Loop OS resolves the fundamental conflict between deep generative latent prediction and deterministic safety. By combining slow-track MCR^2 orthogonal representation geometry with fast-track $O(1)$ pre-sintered barrier filtering, the system guarantees discrete-time forward invariance ($B(S_t) \ge \epsilon$) across continuous physical dynamics in $1.08\,\mu\mathrm{s}$.

Open-Source Transparency: All empirical metrics can be independently reproduced in 1.10 ms via `python reproduce_all.py`. Repositories and 1-click Colab notebooks are publicly hosted at https://github.com/SDRmsung/DLOS and https://github.com/SDRmsung/PCOS under BSL 1.1 (Engine) and CC BY 4.0 (Manuscript).